Izzy Silver

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Computer Engineering student at the University of Waterloo (transferred from Architectural Engineering) who turns LiDAR scans and BIM models into real-time 3D experiences and is building interactive 3D web projects with Three.js.

EDUCATION

University of Waterloo | B.A.Sc. Computer Engineering (Co-op), expected 2030
- President's Scholarship of Distinction (2025)
- Relevant coursework: Linear Algebra, Algorithms & Data Structures (in progress), Numerical Methods (in progress), Digital Circuits and Systems

Gray Academy | High School Valedictorian, 2024

TECHNICAL SKILLS

- **Real-Time 3D:** Unreal Engine 5 (Blueprints, UMG), Datasmith, Dataprep
- **3D Data & Pipelines:** Leica Cyclone 3DR (point clouds, scan-to-mesh), photogrammetry, Revit to Unreal workflows, FBX
- **Languages:** C++, Python, Verilog, RISC-V; familiar with JavaScript, HTML/CSS and Three.js (through AI-assisted projects)
- **Tools:** Git, Perforce, WordPress, Arduino

WORK EXPERIENCE

Studio Lab XR | Architectural Visualisation Researcher | Winnipeg, MB | Jan 2025 – Apr 2025
- Built a VR experience of *The Wall* by Bruce Head (Canada's largest concrete relief) in Unreal Engine 5 from LiDAR data, in collaboration with Prof. Jason Shields (University of Manitoba).
- Converted point clouds to real-time-ready meshes in Leica Cyclone 3DR (cleaning, scan-to-mesh, hole-filling), evaluating Q3 and Q4 scans to balance fidelity against file size.
- Built interactive Blueprint/UMG features: a VR menu with lighting and media controls, dynamic information panels with matching lighting, and grab interactions.
- Documented the Revit, Datasmith, Dataprep and Unreal pipeline (material substitution, metadata filtering, mesh and UV fixes) for architecture firms and students.

QDoc Inc. | Website Developer | Winnipeg, MB | Jun 2021 – Jun 2024
- Designed, built and maintained the marketing website for QDoc's healthcare app, increasing application awareness and usage.
- Helped define QDoc's brand identity, including typography and colour palette, which are still in use today.
- Helped oversee and support co-op students from Red River College Polytech's Web Development program.

B'nai Brith Wilderness Camp | Program Coordinator | Winnipeg, MB | May 2026 – Aug 2026
- Designed and ran camp-wide programs, including a scavenger-hunt challenge trail, a values-based Capture the Flag game, and a staff murder-mystery evening for about 51 staff with full casting, character cards and scripts.
- Produced print-ready program documents and run sheets for multiple age divisions to consistent formatting standards, and built an Excel scoring system for Survivor Day.
- Coordinated camp operations: scaled food preparation for up to about 250 people, designed a parent email management system for the camp office, and drafted supply purchase requests.

PROJECTS

Interactive 3D Portfolio | Three.js, JavaScript, Vite
- Designed and built, with AI-assisted development, a real-time 3D portfolio in Three.js and vanilla JavaScript featuring a photogrammetry scan of myself as the hero scene, floating navigation labels and animated page transitions.
- Check it out here: isilver.uwce.ca

LEADERSHIP

Engineering Society, University of Waterloo | CFO and Academic Rep (elected) | Sep 2026 – Dec 2026
- Elected to multiple Engineering Society positions, most recently CFO and Academic Rep; previously Class Rep (2024).